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An Internship Report

On

“Machine Learning with Python”

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Submitted in partial fulfilment of requirements for the award of the degree of

Bachelor of Engineering

in

Electronics and Communication Engineering

Submitted by

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

VEMANA INSTITUTE OF TECHNOLOGY

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Karnataka Reddy Jana Sangha®

VEMANA INSTITUTE OF TECHNOLOGY

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Koramangala, Bengaluru-560034.



DEPARTMENT OF
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CERTIFICATE

This is to certify that the Internship “**Karunadu Technology Private Limited**” is a bonafide work carried out by **Mr. SHRIHARSHA bearing 1VI18EC123** during the academic year 2024-25 in partial fulfilment of the requirement for the award of **Bachelor of Engineering in Electronics and Communication Engineering** of the **Visvesvaraya Technological University, Belagavi**.

It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The internship report has been approved as it satisfies the academic requirements in respect of the Internship/ Professional Practice prescribed for the said degree.

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Abstract

Machine Learning (ML) is the scientific study of algorithms and statistical models that computer systems use to perform a specific task without being explicitly programmed. Learning algorithms in many applications that we make use of daily. Every time a web search engine like Google is used to search the internet, one of the reasons that work so well is because a learning algorithm to rank web pages. These algorithms are used for various purposes like data mining, image processing, predictive analytics, etc. to name a few. The main advantage of using machine learning is that, once an algorithm learns what to do with data, it can do its work automatically. The brief review and future prospect of the vast applications of machine learning algorithms has been made. Nowadays, large amount of data is available everywhere. Therefore, it is very important to analyse this data in order to extract some useful information and to develop an algorithm based on analysis. This can be achieved through data mining and machine learning. Machine learning is an integral part of artificial intelligence, which is used to design algorithms based on the data trends and historical relationships between data. Machine learning is used in various fields such as bioinformatics, intrusion detection, Information retrieval, game playing, marketing, malware detection, image deconvolution and so on. The certificate would be valid for a period of 3 years dating from the day of issue.

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Chapter 1

INTRODUCTION TO PYTHON

Python is a general-purpose high-level programming language. Python was developed by Guido Van Rossem in 1989 while working at National Research Institute at Netherlands. But officially Python was made available to public in 1991. The official Date of Birth for Python is: Feb 20th 1991. Python is recommended as first programming language for beginners.

Ex: In C:

```
#include<stdio.h>
void main ()
{
    print ("Hello world");
}
```

Output: Hello world

Ex: In Python:

```
Print ("Hello world");
Output: Hello world
```

Python can be used:

The most common important application areas are:

1. Web Development

It is one of the most astonishing applications of Python. Python has a wide range of frame works like Django, Flask, Bottle, and a lot more that provide ease to developers. Furthermore, Python has inbuilt libraries and tools.

2. Game Development

Python comes loaded with many useful extensions (libraries) that come in handy development of interactive games. For instance, libraries like pysoy and PyGame.

1.1 Features of Python:

1.1.1 Simple and easy to learn:

- Python is a simple programming language. When Python program is read, User can feel like reading English statements.

- The syntaxes are very simple and only 30+ keywords are available.
- When compared with other languages, User can write programs with very smaller number of lines. Hence more readability and simplicity.
- User can reduce development and cost of the project.

1.1.2. Freeware and Open Source:

- Python software without any license and it is freeware.
- Its source code is open, so that it can customize based on our requirement.
- **Ex:** Python is customized version of Python to work with Java applications.

1.1.3. High Level Programming language:

- Python is high level programming language and hence it is programmer friendly language.
- Being a programmer, it is not required to concentrate low level activities like memory management and security etc.

1.1.4. Platform Independent:

- Once a Python program is written, it can run on any platform without rewriting once again.
- Internally PVM (Python Virtual Machine) is responsible to convert into machine understandable form.

1.1.5. Portability:

Python programs are portable i.e., it can migrate from one platform to another platform very easily. Python programs will provide same results on any platform.

1.1.6. Dynamically Typed:

- In Python it's not required to declare type for variables. Whenever they are assigning the value, based on value, type will be allocated automatically. Hence Python is considered as dynamically typed language.
- But Java, C etc. are Statically Typed Languages because that have to provide type at the beginning only.
- This dynamic typing nature will provide more flexibility to the programmer.

1.1.7. Both Procedure Oriented and Object Oriented:

Python language supports both procedures oriented (like C, Pascal etc) and object oriented (Like C++, Java) features. Hence, it can get benefits of both like security and reusability etc.

1.1.8. Interpreted:

It is not required to compile Python programs explicitly. Internally Python interpreter will take care of that compilation. If compilation fails then interpreter will raise syntax error.

1.1.9. Extensible:

- Other language programs can be used in Python. The main advantages of this approach are:
- Already existing legacy non-Python code can be used.
- It can improve performance of the application.

1.1.10. Embedded:

Python programs can be used in any other language programs. That is, it can embed Python Programs anywhere.

1.1.11. Extensive Library:

Python has a rich inbuilt library. Being a programmer, User can use this library directly and not responsible to implement the functionality.

1.2 Limitations of Python:

1. Performance wise not up to the mark because it is interpreted language.
2. Not using for mobile applications.

1.3 Identifiers:

A name in Python program is called identifier. It can be class name or function name or module name or variable name.

Rules to define identifiers in Python:

The only allowed characters in Python are

- alphabet symbols (either lower case or upper case)
- digits (0 to 9)
- underscore symbol ()

By mistake if I used any other symbol like \$ then we will get syntax error.

- cash = 10 (correct)
- cash =20 (wrong)

1. Identifier should not start with digit

- 123total (wrong)

- `total123(correct)`

2. Identifiers are case sensitive. of course, Python language is case sensitive language.

- `total=10`
- `TOTAL=999`
- `print(total) #10`
- `print (TOTAL) #999`

1.4 Reserved Words:

In Python some words are reserved to represent some meaning or functionality. Such type of words is called reserved words. There are 33 reserved words available in Python.

Note:

All Reserved words in Python contain only alphabet symbols. Except the following 3 reserved words, all contain only lower-case alphabet symbols.

- `True`
- `False`
- `None`

Example: `a= true (wrong)`

`a=True (correct)`

To check the reserve words in python:

```
>>> import keyword
```

```
>>> keyword
```

```
['False', 'None', 'True', 'and', 'as', 'assert', 'break', 'class', 'continue', 'def', 'del', 'elif', 'else',  
'except', 'finally', 'for', 'from', 'global', 'if', 'import', 'in', 'is', 'lambda', 'nonlocal', 'not', 'or',  
'pass', 'raise', 'return', 'try', 'while', 'with', 'yield']
```

1.5 Data Types:

Data Type represents the type of data present inside a variable. In Python are not required to specify the type explicitly. Based on value provided, the type will be assigned automatically. Hence Python is Dynamically Typed Language. Python contains the following inbuilt data types are int, float, complex, bool, str, bytes, byte array, range, list, tuple, set.

Note: Python contains several inbuilt functions

- **type ():** To check the type of variable
- **id ():** To get address of object
- **print ():** To print the value

int data type:

int data type is used to represent whole numbers (integral values)

Ex: a=10
 type(a) #int

Note:

Python has long data type to represent very large integral values. But in Python there is no long type explicitly and it can represent long values also by using int type only. int values can be represented in the following ways:

1. Decimal form
2. Binary form
3. Octal form
4. Hexa-decimal form

1.5.1. Decimal form(base-10):

It is the default number system in Python

The allowed digits are: 0 to 9

Ex: a =10

1.5.2. Binary form (Base-2):

The allowed digits are: 0 & 1

Literal value should be prefixed with 0b or 0B

Ex: a = 0B1111
 a =0B123
 a=b111

1.5.3. Octal Form (Base-8):

The allowed digits are: 0 to 7

Literal value should be prefixed with 0o or 0O.

Ex: a=0o123
 a=0o786

The allowed digits are: 0 to 9, a-f (both lower and upper cases are allowed)
Literal value should be prefixed with 0x or 0X

Ex: a=0XFACE

a=0XBeef

a=0XBeer

Note: Being a programmer, User can specify literal values in decimal, binary, octal and hexadecimal forms. But PVM will always provide values only in decimal form.

```
a=10
```

```
b=0o10
```

```
c=0X10
```

```
d=0B10
```

```
print(a) >> 10
```

```
print(b) >> 8
```

```
print(c) >> 16
```

```
print(d) >> 2
```

1.6 Base Conversions:

Python provide the following in-built functions for base conversions.

1.6.1. bin ():

bin () can be used to convert from any base to binary.

Ex:1 print (bin (15)) >> '0b1111'

1.6.2. oct ():

oct () can be used to convert from any base to octal

Ex:1 oct (10) >> '0012'

1.6.3. hex ():

hex () can be used to convert from any base to hexa-decimal

Ex:1 >>> hex (100) >> '0x64'

Float data type:

float data type can be used to represent floating point values (decimal values)

Ex: f=1.234

```
type(f) >> float
```

Also represent floating point values by using exponential form (scientific notation)

Ex: `f=1.2e3`

```
print(f) >> 1200.0
```

The main advantage of exponential form is, programmer can represent big values in less memory.

Note:

int values can be represented in decimal, binary, octal and hexa-decimal forms and float values only by using decimal form.

Ex:1) `f=0B11.01`

- 2) File "<stdin>", line 1
- 3) `f=0B11.01 >> Syntax Error: invalid syntax`
- 4) `f=0o123.456 >> Syntax Error: invalid`
- 5) `f=0X123.45 >> Syntax Error: invalid syntax`

Complex Data Type:

A Complex number is of the form a and b contain integers or floating-point values.

Ex: `3+5j`

in the real part, User can use int value and then can specify that either by decimal, octal, binary or hexa-decimal form. But imaginary part should be specified only by using decimal form.

- 1) `a=0B11+5j`
- 2) `print(a) >> 3+5j`
- 3) `a=3+0B11j >> Syntax Error: invalid syntax`

Even can perform operations on complex type values.

- 1) `a=10+1.5j, b=20+2.5j`
- 2) `c=a + b`
- 3) `print(c) >> 30+4j`
- 4) `type(c) >> <class 'complex'>`

Note: Complex data type has some inbuilt attributes to retrieve the real part and imaginary part.

```
c=10.5+3.6j  
c.real==>10.5
```

The complex type generally in scientific applications and electrical engineering applications.

bool data type:

bool data type can be used to represent Boolean values. The only allowed values for this data type are. True and False. Internally Python represents True as 1 and False as 0.

Ex: b=True

```
type(b) >> bool
```

Ex:

```
a=10
b=20
c=a<b

print(c) >> True
True+True >> 2
True-False >> 1
```

str type:

‘str’ represents String data type. A String is a sequence of characters enclosed within single quotes or double quotes.

```
s1='Karunadu'
s1="Karunadu"
```

By using single quotes or double quotes, programmer cannot represent multi line string literals. s1="Karunadu Technologies pvt ltd"

For this requirement, User should go for triple single quotes (""") or triple double quotes (""")

```
s1="""Karunadu
tech pvt ltd"""
s1="""Karunadu
tech pvt ltd"""
```

User can also use triple quotes to use single quote or double quote in ourString. "

```
This is " character"
' This i " Character '
```

User can embed one string in another string.

```
"""This "Python class very helpful" for java students"
```

Slicing of Strings:

Slice means a piece, [] operator is called slice operator, which can be used to retrieve parts of String. In Python Strings follows zero based index.

The index can be either +ve or -ve.

+ve index means forward direction from Left to Right.

-ve index means backward direction from Right to Left.

- 1) s="Karunadu"
- 2) print (s [0]) >> 'K'
- 3) print (s [1]) >> 'a'
- 4) print (s [-1]) >> 'u'
- 5) print (s [40]) >> Index Error: string index out of range

1.7 Python Operators

The operator can be defined as a symbol which is responsible for a particular operation between two operands. Operators are the pillars of a program on which the logic is built in a particular programming language. Python provides a variety of operators described as follows.

1. Arithmetic Operators
2. Relational Operators or Comparison Operators
3. Logical Operators
4. Bitwise Operators
5. Assignment Operators
6. Special operators

Python Comments:

Comments in Python can be used to explain any program code. It can also be used to hide the code as well. Comments are the most helpful stuff of any program. It enables us to understand the way, a program works. In python, any statement written along with # symbol is known as a comment. The interpreter does not interpret the comment.

Comment is not a part of the program, but it enhances the interactivity of the program and makes the program readable.

Python supports two types of comments:

1) Single Line Comment:

In case user wants to specify a single line comment, then comment must start with '#'.
‘#’.

Ex: # This is single line comment.


```
print "Hello Python"
```

Output: Hello Python

2) Multi Line Comment:

Multi lined comment can be given inside triple quotes.

Ex:

```
""" This Is Multiline comment"""
```

```
#Single line comment
```

```
print "Hello Python"
```

Ex:

```
"""This is multiline comment"""
```

Output: Hello Python

1.8 Conditional Statements:

The if statement is used to test a particular condition and if the condition is true, it executes a block of code known as if-block, if the condition is false, it will come out of the loop. The condition of if statement can be any valid logical expression which can be either evaluated to true or false. Figure 1.1 shows the Usage of if condition.

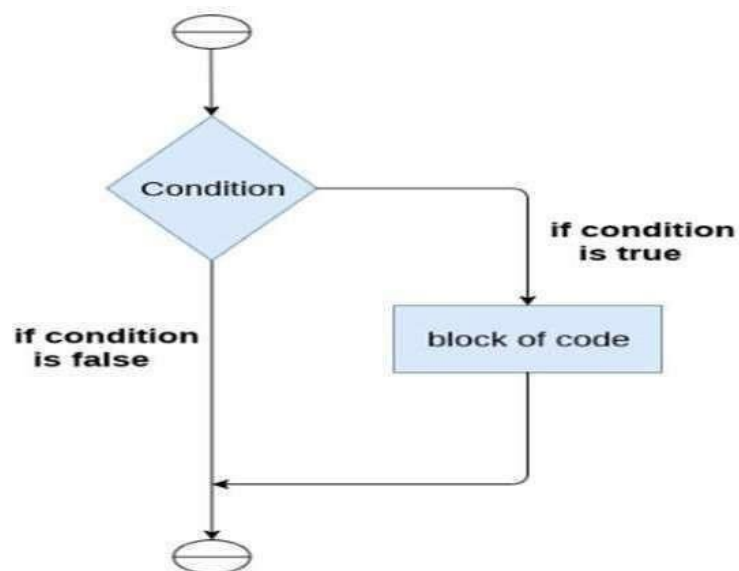


Figure 1.1: Usage of if condition

Syntax:

```
if condition: statement
```

or

```
if condition:  
statement-1
```

statement-2

If condition is true then statements will be executed.

The if-else statement:

The if-else statement provides an else block combined with the if statement which is executed in the false case of the condition. If the condition is true, then the if-block is executed. Otherwise, the else-block is executed. Figure 1.2 shows the Usage of if-else condition.

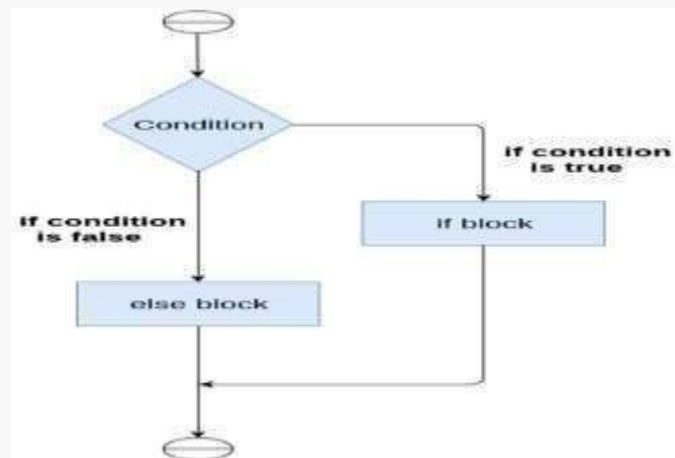


Figure 1.2: Usage of if-else condition

The **syntax** of the if-else statement is given below.

if condition:

 #Block of statements

else:

 #Another block of statements (else-block)

The elif statement enables us to check multiple conditions and execute the specific block of statements depending upon the true condition among them. There can be any number of elif statements in our program depending upon our need. However, using elif is optional. The elif statement works like an if-else-if ladder statement in C. It must be succeeded by an if statement.

The **Syntax** of the elif statement is given below.

if expression 1:

 # Block of statements

elif expression 2:

 # Block of statements

elif expression 3:

 # Block of statements

else:

```
# Block of statements
```

Figure 1.3 shows the Usage of if-elif-else condition.

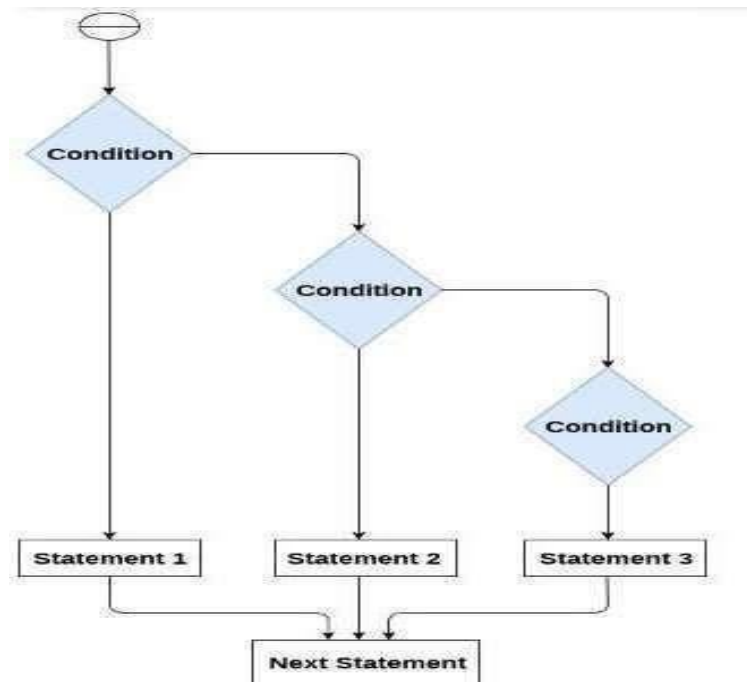


Figure1.3: Usage of if-elif-else condition

1.9 FUNCTIONS:

If a group of statements is repeatedly required then it is not recommended to write these statements every time separately. User have to define these statements as a single unit and can call that unit any number of times based on our requirement without rewriting. This unit is nothing but function. The main advantage of functions is code Reusability.

Note: In other languages functions are known as methods, procedures, subroutines etc. Python supports 2 types of functions

1. Built in Functions
2. er Defined Functions

1. Built in Functions:

The functions which are coming along with Python software automatically, are called built in functions or pre-defined functions.

Ex:

```
id ()  
type ()  
input()  
eval ()
```

2. User Defined Functions:

The functions which are developed by programmer explicitly according to business requirements are called user defined functions.

Syntax to create user defined functions:

```
def function name(parameters):  
    "doc string"
```

```
-----
```

```
    return value
```

Exception Handling:

In any programming language there are 2 types of possible errors

1. Syntax Errors
2. Runtime Errors

1. Syntax Errors:

The errors which occur because of invalid syntax are called syntax errors.

Ex1:

```
x=10  
  
if      x==10  
print("Hello")  
Syntax Error: invalid syntax
```

Ex 2: print "Hello"

Syntax Error: Missing parentheses in call to 'print'

Note:

Programmer is responsible to correct these syntax errors. Once all syntax errors are corrected then only program execution will be started.

2. Runtime Errors:

These are also known as exceptions. While executing the program if something goes wrong because of end user input or programming logic or memory problems and then will get Runtime Errors.

Ex:

```
print (10/0) ==> Zero Division Error: division by zero  
  
print(10/"ten") ==> Type Error: unsupported operand type(s) for /: 'int' and 'str'
```

```
x=int (input ("Enter Number:"))  
print(x)  
D:\Python_classes>py test.py  
Enter Number: ten  
Value Error: invalid literal for int () with base 10: 'ten'
```

Note: Exception Handling concept applicable for Runtime Errors but not for syntax errors.

Exception Handling

An unwanted and unexpected event that disturbs normal flow of program is called exception handling.

Ex:

Zero Division Error

Type Error

Value Error

File Not Found Error

It is highly recommended to handle exceptions. The main objective of exception handling is Graceful Termination of the program (i.e., User should not block our resources and Programmer should not miss anything) Exception handling does not mean repairing exception. User have to define alternative way to continue rest of the program normally. For example, our programming requirement is reading data from remote file locating at London. At runtime if London file is not available then the program should not be terminated abnormally. User have to provide local file to continue rest of the program normally. This way of defining alternative is nothing but exception handling.

try:

read data from remote file locating at London

except File Not Found Error:

use local file and continue rest of the program normally

Default Exception Handling in Python:

Every exception in Python is an object. For every exception type the corresponding classes are available. Whenever an exception occurs PVM (Python Virtual Machine) will create the corresponding exception object and will check for respective handling code. If handling code is not available then Python interpreter terminates the program abnormally and prints corresponding exception information to the console.

The rest of the program won't be executed.

Ex:

- 1) `print("Hello")`
- 2) `print (10/0)`
- 3) `print("Hi")`
- 4) `D:\Python_classes>py test.py`
- 5) Hello
- 6) Traceback (most recent call last):
- 7) File "test.py", line 2, in <module>
- 8) `print (10/0)`
- 9) Zero Division Error: division by zero

CHAPTER 2

ORGANIZATION PROFILE

Karunadu Technologies Private Limited, is a well-established and experienced IT software solutions and services industry focusing on quality standards and customer values. The company offer broad range of customized software applications powered by concrete technology and industry expertise. Karunadu Technologies private limited offers end to end embedded solutions to its customers that entail all stages of product life cycle. Our proficient and dedicated embedded application development team helps you to build embedded application solutions that will ensure the best customer satisfaction. The customer requirements are matched with our strong experience and in-depth expertise of the market. Karunadu Technologies private limited is also a leading Skills and Talent Development company that is building a manpower pool for global industry requirements. Empowers individual with knowledge, skills and competencies that assist them to escalates integrated individuals with a sense of commitment and dedication towards the society. Karunadu Technologies private limited also helps companies to find right individuals matching the requirements. Company is engaged in Outsourcing of talented candidates also find you the best talent of pre-screened, interested candidates to your recruiters and/or hiring managers. Karunadu Technologies private limited helps students to transform their own ideas to project. Assistance to individuals for their IEEE Academic project are provided. Firm had got large collection of projects on latest technologies. World class innovative ideas on Embedded, IOT, Mat lab, VLSI, Android, Big Data, Dot net, Java and much more are developed.

VISION:

To Empower Unskilled Individual with knowledge, skills and technical competencies in the field of Information Technology and Embedded engineering which assist them to escalate as integrated individuals contributing to company's and Nation's growth.

Mission:

Provide cost effective and reliable solutions to customers across various latest technologies., Offer scalable end-to-end application development and management

solutions, Provide cost effective highly scalable products for varied verticals. , Focus on creating sustainable value growth through innovative solutions and unique partnerships. , Create, design and deliver business solutions with high value and innovation by leveraging technology expertise and innovative business models to address long-term business objectives. , Keep our products and services updated with the latest innovations in the respective requirement and technology.

CHAPTER 3

MACHINE LEARNING

Machine Learning is a system that can learn from example through self-improvement and without being explicitly coded by programmer. The breakthrough comes with the idea that a machine can singularly learn from the data (i.e., example) to produce accurate results.

Machine learning combines data with statistical tools to predict an output. This output is then used by corporate to makes actionable insights. Machine learning is closely related to data mining and Bayesian predictive modelling. The machine receives data as input, use an algorithm to formulate answers.

A typical machine learning tasks are to provide a recommendation. For those who have a Netflix account, all recommendations of movies or series are based on the user's historical data. Tech companies are using unsupervised learning to improve the user experience with personalizing recommendation.

Machine learning is also used for a variety of task like fraud detection, predictive maintenance, portfolio optimization, automatize task and so on.

Machine Learning vs. Traditional Programming

Traditional programming differs significantly from machine learning. In traditional programming, a programmer codes all the rules in consultation with an expert in the industry for which software is being developed. Each rule is based on a logical foundation; the machine will execute an output following the logical statement. When the system grows complex, more rules need to be written. It can quickly become unsustainable to maintain.

Machine learning is supposed to overcome this issue. The machine learns how the input and output data are correlated and it writes a rule. The programmers do not need to write new rules each time there is new data. The algorithms adapt in response to new data and experiences to improve efficacy over time.

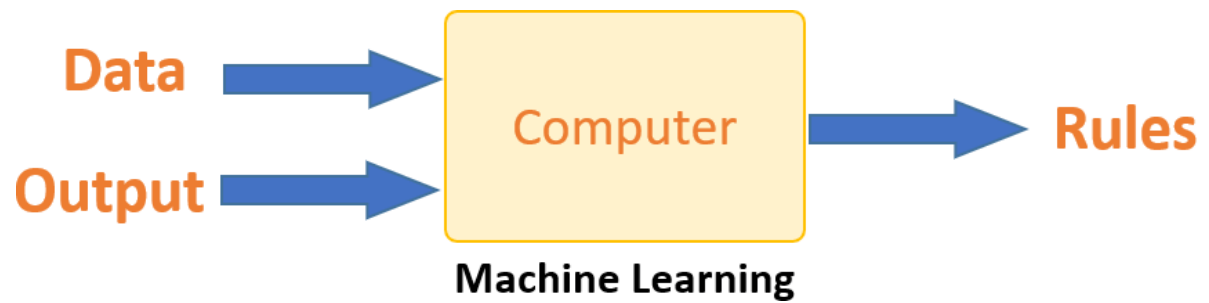


Figure 3.1: Block Diagram of Machine Learning

Machine learning:

Machine learning is the brain where all the learning takes place. The way the machine learns is similar to the human being. Humans learn from experience. The more we know, the more easily it can predict. By analogy, when programmer face an unknown situation, the likelihood of success is lower than the known situation. Machines are trained the same. To make an accurate prediction, the machine sees an example. When User give the machine a similar example, it can figure out the outcome. However, like a human, if it's feed a previously unseen example, the machine has difficulties to predict.

The core objective of machine learning is the learning and inference. First of all, the machine learns through the discovery of patterns. This discovery is made thanks to the data. One crucial part of the data scientist is to choose carefully which data to provide to the machine. The list of attributes used to solve a problem is called a feature vector and canthink of a feature vector as a subset of data that is used to tackle a problem.

The machine uses some fancy algorithms to simplify the reality and transform this discovery into a model. Therefore, the learning stage is used to describe the data and summarize it into a model.

Machine Learning is Required for

The world today is evolving and so are the needs and requirements of people. Furthermore, User are witnessing a fourth industrial revolution of data. In order to derive meaningful insights from this data and learn from the way in which people and the system interface with the data, Programmer need computational algorithms that can mix the data and provide us with results that would benefit us in various ways. Machine Learning has revolutionized

industries like medicine, healthcare, manufacturing, banking, and several other industries. Therefore, Machine Learning has become an essential part of modern industry.

Data is expanding exponentially and in order to harness the power of this data, added by the massive increase in computation power, Machine Learning has added another dimension to the way User perceive information. Machine Learning is being utilized everywhere. The electronic devices use, the applications that are part of your everyday life are powered by powerful machine learning algorithms.

Machine learning example – Google is able to provide us with appropriate search results based on browsing habits. Similarly, Netflix is capable of recommending the films or shows that it wants to watch based on the machine learning algorithms that perform predictions based on your watch history.

Furthermore, machine learning has facilitated the automation of redundant tasks that have taken away the need for manual labor. All of this is possible due to the massive amount of data that generate on a daily basis. Machine Learning facilitates several methodologies to makesense of this data and provide us with steadfast and accurate results.

3.1 Types of Machine Learning

3.1.1 Supervised Machine Learning

Supervised learning algorithms are used when the output is classified or labelled. These algorithms learn from the past data that is inputted, called as training data, runs its analysis and uses this analysis to predict future events of any new data within the known classifications. The accurate prediction of test data requires large data to have a sufficient understanding of the patterns. The algorithm can be trained further by comparing the training outputs to actual ones and using the errors for modification of the algorithms.

Real Life Example:

Image Classification – The algorithm is drawn from feeding with labelled image data. An algorithm is trained and it is expected that in the case of the new image the algorithm classifies it correctly.

Market Prediction – It is also called Regression. Historical business market data is fed to the computer. With analysis and regression algorithm new price for the future is predicted depending on variables.

3.1.2. Unsupervised Machine Learning

Unsupervised learning algorithms are used when we are unaware of the final outputs and the classification or labelled outputs are not at our disposal. These algorithms studies and generate a function to describe completely hidden and unlabeled patterns. Hence, there is no correct output, but it studies the data to give out unknown structures in unlabeled data.

Real Life Example:

Clustering – Data with similar traits are asked to group together by the algorithm, this grouping is called clusters. These prove helpful in the study of these groups which can be applied on the entire data within a cluster more or less.

High Dimension Data – High dimension data is normally not easy to work with. With the help of unsupervised learning, visualization of high dimension data becomes possible.

Generative Models – Once your algorithm analyses and comes up with the probability distribution of the input, it can be used to generate new data. This proves to be very helpful in cases of missing data.

3.1.3. Reinforcement Machine Learning

This type of machine learning algorithm uses the trial-and-error method to churn out output based on the highest efficiency of the function. The output is compared to find out errors and feedback which are fed back to the system to improve or maximize its performance. The model is provided with rewards which are basically feedback and punishments in its operations while performing a particular goal.

3.1.4. Semi-Supervised Machine Learning

These algorithms normally undertake labelled and unlabeled data, where the unlabeled data amount is large as compared to labelled data. As it works with both and in between supervised and unsupervised learning algorithms, therefore is called semi-supervised machine learning. Systems using these models are seen to have improved learning accuracy.

Example – An image archive can contain only some of its data labelled, e.g., Dog, cat, mouse, and a large chunk of images remain unlabeled.

3.2 Models based on the Kind of Outputs from the Algorithms

There are two types of machine learning algorithms, they are

- Supervised machine learning
- Unsupervised machine learning

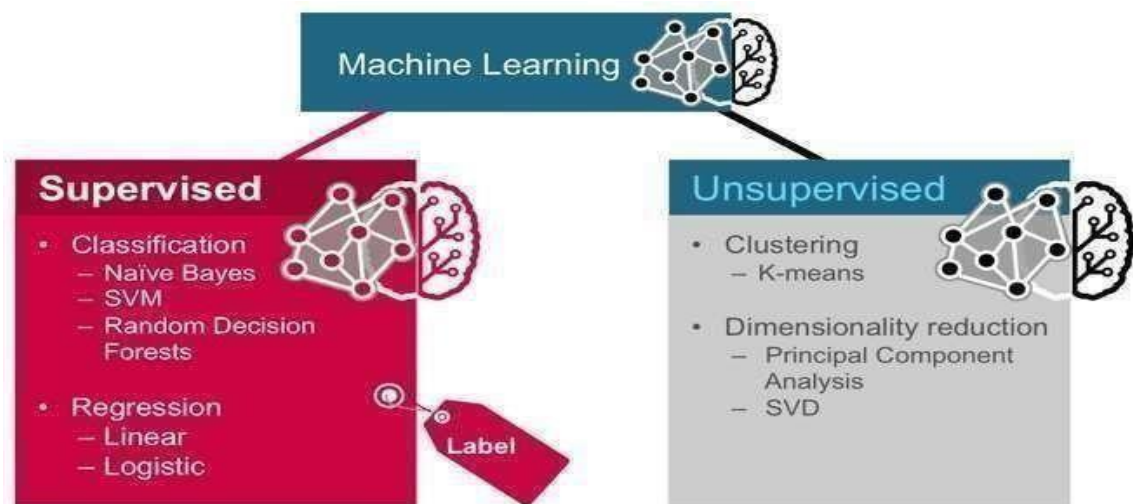


Figure 3.2: Machine learning Types

Supervised learning:

Supervised learning as the name indicates the presence of a supervisor as a teacher. Basically, supervised learning is a learning in which we teach or train the machine using data which is well labeled that means some data is already tagged with the correct answer. After that, the machine is provided with a new set of examples(data) so that supervised learning algorithm analyses the training data (set of training examples) and produces a correct outcome from labeled data.

For instance, suppose a basket is filled with different kinds of fruits. Now the first step is to train the machine with all different fruits one by one like this:

If shape of object is rounded and depression at top having color Red then it will be labelled as –Apple.

If shape of object is long curving cylinder having color Green-Yellow then it will be labelled as –Banana.

Since the machine has already learned the things from previous data and this time have to use it wisely. It will first classify the fruit with its shape and color and would confirm the fruit name as banana and put it in Banana category. Thus, the machine learns

the things from training data (basket containing fruits) and then apply the knowledge to test data (new fruit).

Supervised learning classified into two categories of algorithms:

Classification: A classification problem is when the output variable is a category, such as “Red” or “blue” or “disease” and “no disease”.

Regression: A regression problem is when the output variable is a real value, such as “dollars” or “weight”.

Challenges in Supervised machine learning

- Irrelevant input feature present training data could give inaccurate results. Data preparation and pre-processing is always a challenge. Accuracy suffers when impossible, unlikely, and incomplete values have been inputted as training data
- If the concerned expert is not available, then the other approach is "brute-force." It means to think that the right features (input variables) to train the machine on. It could be inaccurate.

Advantages of Supervised Learning

- Supervised learning allows us to collect data or produce a data output from the previous experience.
- Helps us to optimize performance criteria using experience Supervised machine learning helps to solve a various types of real-world computation problems.

Disadvantages of Supervised Learning

- Decision boundary might be overstrained if your training set which doesn't have examples that are want to have in a class.
- User need to select lots of good examples from each class while training the classifier.
- Classifying big data can be a real challenge. Training for supervised learning needs a lot of computation time.

3.3 KNN (K Nearest Neighbors)

KNN Algorithm

K nearest neighbors or KNN Algorithm is a simple algorithm which uses the entire dataset in its training phase. Whenever a prediction is required for an unseen data instance, it searches through the entire training dataset for k-most similar instances and the data with the most similar instance is finally returned as the prediction.

Features Of KNN Algorithm

The KNN algorithm has the following features:

- KNN is a Supervised Learning algorithm that uses labeled input data set to predict the output of the data points.
- It is one of the simplest Machine learning algorithms and it can be easily implemented for a varied set of problems.
- It is mainly based on feature similarity. KNN checks how similar a data point is to its neighbor and classifies the data point into the class it is most similar to.
- Unlike most algorithms, KNN is a non-parametric model which means that it does not make any assumptions about the data set. This makes the algorithm more effective since it can handle realistic data.
- KNN is a lazy algorithm, this means that it memorizes the training data set instead of learning a discriminative function from the training data.
- KNN can be used for solving both classification and regression problems.

Working of KNN algorithm

In KNN, K is the number of nearest neighbors. The number of neighbors is the core deciding factor. K is generally an odd number if the number of classes is 2. When K=1, then the algorithm is known as the nearest neighbor algorithm. Suppose P1 is the point, for which label needs to predict. First, Programmer need to find the one closest point to P1 and then the label of the nearest point assigned to P1.

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Suppose P1 is the point, for which label needs to predict. First, User need to find the k closest point to P1 and then classify points by majority vote of its k neighbors. Each object votes for their class and the class with the most votes is taken as the prediction. For finding closest similar points and find the distance between points using distance measures such as Euclidean distance, Hamming distance, Manhattan distance and Makowski distance. KNN has the following basic steps:

3.4 Linear Regression

It is one of the most widely known modelling technique. Linear regression is usually among the first few topics which people pick while learning predictive modelling. In this technique, the dependent variable is continuous, independent variable(s) can be continuous or discrete, and nature of regression line is linear.

Linear Regression establishes a relationship between dependent variable (Y) and one or more independent variables (X) using a best fit straight line (also known as regression line).

It is represented by an equation $Y = a + b \cdot X + e$, where **a** is intercept, **b** is slope of the line and **e** is error term. This equation can be used to predict the value of target variable based on given predictor variable(s).

The difference between simple linear regression and multiple linear regression is that, multiple linear regression has (>1) independent variables, whereas simple linear regression has only 1 independent variable. Now, the question is “How can be obtain best fit line?”.

How to obtain best fit line (Value of a and b)?

This task can be easily accomplished by Least Square Method. It is the most common method used for fitting a regression line. It calculates the best-fit line for the observed data by minimizing the sum of the squares of the vertical deviations from each data point to the line. Because the deviations are first squared, when added, there is no cancelling out between positive and negative values. The model performance using the metric R-square. To know more details about these metrics, User can read Model Performance metrics.

Important Points about Linear Regression

- There must be linear relationship between independent and dependent variables.
- Multiple regression suffers from multicollinearity, autocorrelation, heteroskedasticity.

- Linear Regression is very sensitive to Outliers. It can terribly affect the regression line and eventually the forecasted values.
- Multicollinearity can increase the variance of the coefficient estimates and make the estimates very sensitive to minor changes in the model. The result is that the coefficient estimates are unstable.

Advantages of Linear Regression

Linear regression is an extremely simple method. It is very easy and intuitive to use and understand. A person with only the knowledge of high school mathematics can understand and use it. In addition, it works in most of the cases. Even when it doesn't fit the data exactly and can use it to find the nature of the relationship between the two variables.

Disadvantage of Linear Regression

- By its definition, linear regression only models' relationships between dependent and independent variables that are linear. It assumes there is a straight-line relationship between them which is incorrect sometimes. Linear regression is very sensitive to the anomalies in the data (or outliers).
- Take for example most of your data lies in the range 0-10. If due to any reason only one of the data items comes out of the range, say for example 15, this significantly influences the regression coefficients.
- Another disadvantage is to have a number of parameters than the number of samples available then the model starts to model the noise rather than the relationship between the variables.

3.5 Multiple Linear Regression

In many cases, there may be possibilities of dealing with more than one predictor variable for finding out the value of the response variable. Therefore, the simple linear models cannot be utilized as there is a need for undertaking multiple linear regression for analyzing the predictor variables. Using the two explanatory variables and can delineate the equation of multiple linear regression as follows.

Advantages of multiple regression

Any disadvantageous used for fitting a regression line. It calculates the best-fit line for the observed data by minimizing the sum of the squares of the vertical deviations from each data point to the line. Because the deviations are first squared, when added, there is no cancelling out between positive and negative values. The model performance using the metric R-square. To know more details about these metrics of using a multiple regression.

the data being used. Two examples of this are using incomplete data and falsely concluding that a correlation is a causation.

When reviewing the price of homes, for example, suppose the real estate agent looked at only 10 homes, seven of which were purchased by young parents. In this case, the relationship between the proximity of schools may lead her to believe that this had an effect on the sale price for all homes being sold in the community. This illustrates the pitfalls of incomplete data. Had she used a larger sample, she could have found that, out of 100 homes sold, only ten percent of the home values were related to a school's proximity. If she had used the buyers' ages as a predictor value, she could have found that younger buyers were willing to pay more for homes in the community than older buyers.

In the example of management salaries, suppose there was one outlier who had a smaller budget, less seniority and with fewer personnel to manage but was making more than anyone else. The HR manager could look at the data and conclude that this individual is being overpaid. However, this conclusion would be erroneous if he didn't take into account that this manager was in charge of the company's website and had a highly coveted skillset in network security.

Disadvantages of multiple regression

- Any disadvantage of using a multiple regression model usually comes down to the data being used. Two examples of this are using incomplete data and falsely concluding that a correlation is a causation.
- When reviewing the price of homes, for example, suppose the real estate agent looked at only 10 homes, seven of which were purchased by young parents. In this case, the relationship between the proximity of schools may lead her to believe that this had an effect on the sale price for all homes being sold in the community. This illustrates the pitfalls of incomplete data. Had she used a larger sample, she could have found that, out of 100 homes sold, only ten percent of the home values were related to a school's proximity. If she had used the buyers' ages as a predictor value, she could have found that younger buyers were willing to pay more for homes in the community than older buyers.

3.6 Polynomial regression

Polynomial regression is a form of regression analysis in which the relationship

between the independent variable x and the dependent variable y is modelled as an n th degree polynomial in x . Polynomial regression fits a nonlinear relationship between the value of x and the corresponding conditional mean of y , denoted $E(y | x)$, and has been used to describe nonlinear phenomena such as the growth rate of tissues, the distribution of carbon isotopes in lake sediments, and the progression of disease epidemics. Although polynomial regression fits a nonlinear model to the data, as a statistical estimation problem it is linear, in the sense that the regression function $E(y | x)$ is linear in the unknown parameters that are estimated from the data. For this reason, polynomial regression is considered to be a special case of Multiple Linear Regression.

PR is given by the formula:



$$y = b_0 + b_1x_1 + b_2x_1^2 + \dots + b_nx_1^n$$

PR (Polynomial Regression) can be used when the SL (Straight Line) doesn't fit well our observations and to obtain parabolic effect.

At the same time PR has its own unique cases, and there is a problem and it might try a SLR and MLR (Multiple Linear Regression) first and see what happens. With a PR sometimes, User obtain better results. For example, PR issued to observe how epidemics spread across population, and similar use cases, so it's a matter of what, User want to predict, and of course it's always good to have more tools in our arsenal, the trick here is that when they discuss about linear and nonlinear models, they are not thinking in terms of variables and the coefficient. So, the question is whether the function can be expressed as a linear combination of these coefficient or not, being ultimately unknown the is goal to find the coefficient values, and this is why linear and nonlinear refers to the coefficients. So, PR is a special case of MLR rather than a standard new type of regression.

Advantages of using Polynomial Regression:

- Polynomial provides the best approximation of the relationship between the dependent and independent variable.

- A Broad range of function can be fit under it.
- Polynomial basically fits a wide range of curvature.

Disadvantages of using Polynomial Regression

- The presence of one or two outliers in the data can seriously affect the results of the nonlinear analysis.
- These are too sensitive to the outliers.
- In addition, there are unfortunately fewer model validation tools for the detection of outliers in nonlinear regression than there are for linear regression.

3.7. Decision Tree

Decision Tree Classification Algorithm

- Decision Tree is a Supervised learning technique that can be used for both classification and Regression problems, but mostly it is preferred for solving Classification problems. It is a tree-structured classifier, where internal nodes represent the features of a dataset, branches represent the decision rules and each leaf node represents the outcome.
- In a Decision tree, there are two nodes, which are the Decision Node and Leaf Node. Decision nodes are used to make any decision and have multiple branches, whereas Leaf nodes are the output of those decisions and do not contain any further branches.
- The decisions or the test are performed on the basis of features of the given dataset.
- It is a graphical representation for getting all the possible solutions to a problem/decision based on given conditions.
- It is called a decision tree because, similar to a tree, it starts with the root node, which expands on further branches and constructs a tree-like structure.
- A decision tree simply asks a question, and based on the answer (Yes/No), it further split the tree into subtrees.
- Decision Tree is a Supervised learning technique that can be used for both classification and Regression problems, but mostly it is preferred for solving Classification problems. It is a tree-structured classifier, where internal nodes represent the features of a dataset, branches represent the decision rules and each leaf node represents the outcome.

Below diagram explains the general structure of a decision tree:

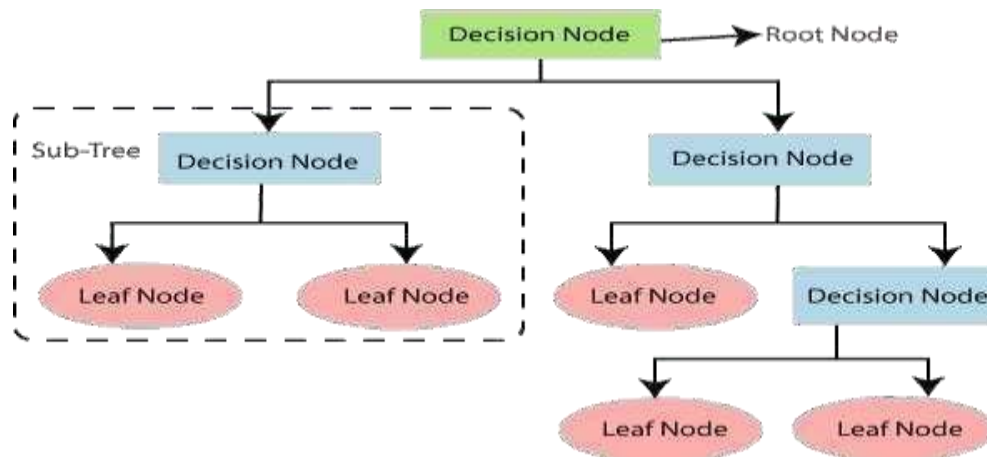


Figure 3.3: General structure of a decision tree

Decision Tree Terminologies

- **Root Node:** Root node is from where the decision tree starts. It represents the entire dataset, which further gets divided into two or more homogeneous sets.
- **Leaf Node:** Leaf nodes are the final output node, and the tree cannot be segregated further after getting a leaf node.
- **Splitting:** Splitting is the process of dividing the decision node/root node into sub-nodes according to the given conditions.
- **Branch/Sub Tree:** A tree formed by splitting the tree.
- **Pruning:** Pruning is the process of removing the unwanted branches from the tree.
- **Parent/Child node:** The root node of the tree is called the parent node, and other nodes are called the child nodes.

How does the Decision Tree algorithm Work?

In a decision tree, for predicting the class of the given dataset, the algorithm starts from the root node of the tree. This algorithm compares the values of root attribute with the record (real dataset) attribute and, based on the comparison, follows the branch and jumps to the next node.

For the next node, the algorithm again compares the attribute value with the other sub-nodes and move further. It continues the process until it reaches the leaf node of the tree. The complete process can be better understood using the below algorithm steps:

Step-1: Begin the tree with the root node, says S, which contains the complete dataset. **Step-2:** Find the best attribute in the dataset using **Attribute Selection Measure (ASM)**. **Step-3:** Divide the S into subsets that contains possible values for the best attributes. **Step-4:** Generate the decision tree node, which contains the best attribute.

Step- 5: Recursively make new decision trees using the subsets of the dataset created in continue this process until a stage is reached where, I cannot further classify the nodes and called the final node as a leaf node.

Example: Suppose there is a candidate who has a job offer and wants to decide whether he should accept the offer or not. So, to solve this problem, the decision tree starts with the root node (Salary attribute by ASM). The root node splits further into the next decision node (distance from the office) and one leaf node based on the corresponding labels. The next decision node further gets split into one decision node (Cab facility) and one leaf node. Finally, the decision node splits into two leaf nodes (Accepted offers and Declined offer). Consider the below diagram:

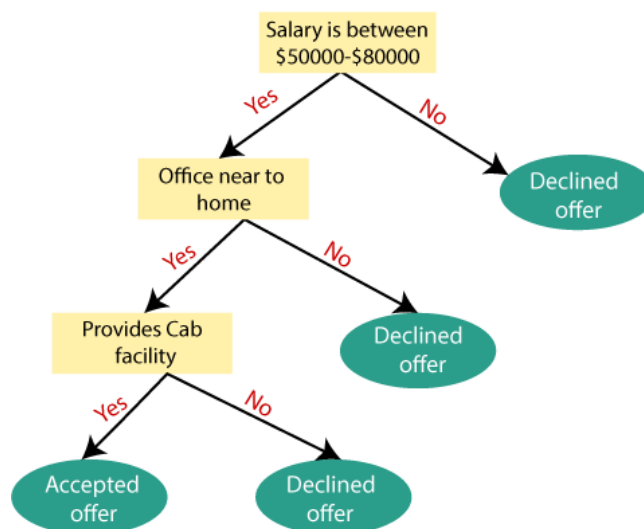


Figure 3.4: Decision tree example

Advantages of the Decision Tree

- It is simple to understand as it follows the same process which a human follow while making any decision in real-life.
- It can be very useful for solving decision-related problems.
- It helps to think about all the possible outcomes for a problem.
- There is less requirement of data cleaning compared to other algorithms.

Disadvantages of the Decision Tree

- The decision tree contains lots of layers, which makes it complex.
- A small change in the data can cause a large change in the structure of the decision tree causing instability.
- For a Decision tree sometimes, calculation can go far more complex compared to other algorithms.
- Decision tree often involves higher time to train the model.
- Decision tree training is relatively expensive as the complexity and time has taken are more.
- The Decision Tree algorithm is inadequate for applying regression and predicting continuous values.

CHAPTER 4

INTERNSHIP PROJECT

Loan Prediction using SVM Algorithm

Support Vector Machine (SVM) is a supervised machine learning algorithm that can be used for both classification or regression challenges. However, it is mostly used in classification problems. In the SVM algorithm, we plot each data item as a point in n-dimensional space (where n is a number of features you have) with the value of each feature being the value of a particular coordinate. Then, we perform classification by finding the hyper-plane that differentiates the two classes very well

Loan Prediction is very helpful for employee of banks as well as for the applicant. The aim of this Paper is to provide quick, immediate and easy way to choose the deserving applicants. It can provide special advantages to the bank. The Loan Prediction System can automatically calculate the weight of each features taking loan processing and on new test data same features are processed with respect to their associated weight. A time limit can be set for the applicant to check whether his/her loan can be sanctioned or not. Loan Prediction System allows jumping to specific application so that it can be checked on priority basis. This Paper is exclusively for the managing authority of Bank/finance company, whole process of prediction is done privately no stakeholders would be able to alter the processing. Result against particular Loan Id can be sent to various department of banks so that they can take appropriate action on application. This helps all other departments to carry out other formalities.

Objectives: To Predict loan is granted or not. Software used: Anaconda, Jupyter notebook

Algorithm used: SVM Algorithm

Data Pre-processing Techniques:

1. Labelling string values using Label Encoder ().

How to do:

- Read the csv file using pandas `OpenCV ()` function.
- Applying Data pre-processing technique for standardizing and scaling of data.
- Divide data set into x and y; x having independents value and y having dependents variables.
- Further dividing x and y into x test, x train and y test, y train respectively using `train testsplit ()`.
- Importing and applying svm algorithm from sk learn import SVM.

CONCLUSION

The company gave a broad spectrum of knowledge in the domain of Machine Learning which required us to learn more about the machine learning. Working on different cross-platforms and different Python libraries will develop my Python Programming skills and Data Analyst skills which in turn helped to acquire knowledge about work-culture of Corporate IT sector.

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